

YIMING CAO

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TECHNICAL SKILLS

Languages	Python, Go, TypeScript / JavaScript, Java, SQL
Backend & APIs	FastAPI, async SQLAlchemy, Alembic, Pydantic, Go Fiber, GORM, Spring Boot, REST API design, JWT auth & RBAC, service-layer architecture
Data Engineering	GeoPandas / Shapely, NetworkX, PostgreSQL / PostGIS, MySQL, MongoDB Atlas, Redis, schema & ERD design, data contracts between services, canonical tables, schema migrations, data validation frameworks
LLM & AI Workflows	HuggingFace, OpenAI API, RAG, vector search, prompt engineering & prefix caching, rule-first LLM pipelines (deterministic filters before model calls), LLM evaluation, Claude Code, spec-driven development (GitHub Spec Kit)
Deep Learning	PyTorch, NumPy, SciPy, CNN backbones, diffusion models, knowledge distillation, few-shot learning, image forensics & quality metrics, Slurm (Compute Canada)
Frontend	React 19, TypeScript, TanStack Query, Redux, React Router, Tailwind CSS, streaming chat UIs
Cloud, DevOps & Testing	Pytest, JUnit / Mockito, Postman, AWS (Elastic Beanstalk, EC2, CloudWatch), Docker Compose, Git / GitHub PR workflow, soak testing, security review

EXPERIENCE

Software Engineer — Tomkulak Consortium (volunteer) Jan 2026 – Present

Critical-infrastructure system-modelling platform: 21 sectors, Vancouver Island MVP scaling to BC, Yukon and Alberta; built in three layers: data ingestion, system modelling, and visualization

- Owned the system-modelling architecture: authored the ERD/object model (nine entities: components, candidate sites, scenario state, dependency links, configuration membership...), the rule separating source GIS fields from model-computed fields, and the versioned data contract between the ingestion layer and the modelling layer — the document the enrichment, configuration-generation and **React** visualization work was then built against, so modelling and frontend proceeded in parallel with ingestion instead of waiting on it.
- Built the dependency and constraint framework: eight relationship categories (direct, cross-sector, parent-child, proximity, corridor, redundancy, exclusion, prerequisite) as typed **NetworkX** edges plus feasibility rules (mandatory / optional / conditionally removable components, prerequisites, minimum redundancy) kept as revisable rules rather than hard-coded functions — the guard that stops the configuration generator from emitting invalid systems.
- Built the multi-objective scoring and ranking engine: six objectives (stakeholder alignment, geostrategic, economic, sustainability, resilience, CI performance) with configurable weightings, separate rankings for 'keep all', 'keep most', and 'allow removals' recommendation types, and explanation metadata naming the factors that drove each rank.
- Built the scenario framework: side-by-side comparison of 3+ configurations, sensitivity analysis over six variables (costs, approval and performance assumptions, weightings, retention constraints), and **Pydantic-typed JSON** result structures matched to the frontend's optimization and topology endpoints, so the frontend switches from mock to live data with one config change.
- Data ingestion layer: engineered the configuration-driven **Python** ingestion template adopted by all 21 sector pipelines (pluggable **GDB / Shapefile / GPKG / ArcGIS** readers, staging → transform → **PostGIS** load) and the validation framework (geometry validity, required fields, uniqueness, referential integrity) whose reports separate code bugs from data problems to send back to the GIS team.

Software Engineer — NextTier, Covina, CA

Sept 2024 – Sept 2025

Designed and built a client RAG chatbot from scratch — backend, retrieval layer, UI, and AWS deployment

- Built the **Java Spring Boot** backend exposing REST APIs for message flow, chat history, session state, and context-aware tone adjustment, using a service-layer pattern over **MongoDB**.
- Implemented the **RAG** retrieval layer — **OpenAI text-embedding-3** queries with **k-NN vector search** on **MongoDB Atlas** — grounding **GPT-4o** responses in the client's verified domain corpus with targeted prompt engineering; enriched the knowledge base from **HuggingFace** datasets.
- Delivered a **React + Tailwind CSS** chat interface with end-to-end token streaming (**OpenAI** → **Spring Boot** → **React**), input throttling, and conversational context retention, iterated over 4 months of client feedback.
- Reached 100% test coverage with **JUnit** and **Mockito**; deployed to **AWS Elastic Beanstalk (EC2)** with **MongoDB Atlas**, autoscaling, and **CloudWatch** monitoring of latency and error rates.

PROJECTS

Job Hunting Assistant: a multi-site job scraper with an LLM matching pipeline

Mar 2026 – Present

Solo · Python, FastAPI, async SQLAlchemy, Alembic, PostgreSQL, Redis, Chrome Extension (MV3), React 19, TypeScript, OpenAI API, Docker Compose

- Designed and built an end-to-end platform that ingests semi-structured data from three heterogeneous web sources into a **FastAPI/PostgreSQL** backend, a **React** console, and a standalone LLM filter-and-match service — 1,306 records ingested in one unattended cycle.

- Designed a dual-write data model — per-source append-only tables plus a 27-column canonical table written atomically — with a projection layer enforcing fixed value sets, source-precedence rules and explicit unknown handling; evolved the schema through 31 **Alembic** migrations.
- Built filter-matcher, a service that encodes expert screening rules as cheap deterministic gates (user preferences, field completeness, hard-requirement filters, deduplication, keyword extraction, and a skills-coverage score) and sends only records with a mid-to-high coverage score to **gpt-4o-mini**; every run must account for every input record (shortlisted, excluded, or flagged) or is recorded as failed — 184 mocked tests.
- Reduced LLM cost by design: ~70% fewer input tokens via one-time condensation of each document to 120–200 words, prompts ordered for the provider's prefix caching, and zero model calls for records rejected by cheaper gates.
- Built an unattended orchestrator in a **Chrome extension** service worker; ran 10-hour soak tests and fixed the race conditions they exposed.
- Practised spec-driven development with **GitHub Spec Kit** and **Claude Code** across 11 feature specs; verifying each feature against live data caught three field-value mismatches (one affecting 28% of records) before release.

E-Commerce Admin Platform — Go REST API and React dashboard

Nov 2025 – Dec 2025

Solo · Go 1.25, Fiber v3, GORM, MySQL, JWT, bcrypt, React 19, TypeScript, Redux, React Router 7

- Built a 28-route **Go REST API (Fiber v3 + GORM/MySQL)** and a 14-route **React/TypeScript** dashboard covering user, role, product and order management, image upload, **CSV** export and a daily-sales analytics chart.
- Implemented cookie-based authentication with HS256 JWTs in HTTP-only cookies, **bcrypt** (cost 14) password hashing, and role-based access control over a many-to-many role-permission model with per-method `view_*/edit_*` permission checks.
- Wrote a generic pagination layer through a **Go Entity** interface shared by users, products and orders, with eager-loaded order items and computed totals to avoid N+1 queries.
- Ported the codebase to current major versions (**Fiber v3, React Router 7, React 19, Redux 5**) and wrote a security review identifying hard-coded secrets, authorisation gaps, unsanitised upload filenames and float-based money as next fixes.

RESEARCH PROJECTS — UNIVERSITY OF BRITISH COLUMBIA

Detecting AI-generated images by their missing Bayer pattern — EECE 541

Apr 2024

- Co-authored a training-free detector: re-mosaic under each of four Bayer layouts, demosaic, and use the spread of PSNRs as the test statistic; 97–100% detection on 46,000 generated images vs 999 camera RAWs, with 90% of real photographs retained.
- Ported **Malvar-He-Cutler** demosaicing and **PSNR/SSIM/CIQE2000** metrics to batched **PyTorch** for GPU throughput; compared six classifiers against a **ResNet-50** baseline and built an in-browser version verified pixel-exact against the **PyTorch** code.

Age-invariant face recognition by knowledge distillation — EECE 571L

Apr 2024

- Distilled a **ResNet-100 ElasticFace** teacher into **ElasticFace** and **MTLFace** students on B3FD, running 48-hour **SLURM** jobs on **Compute Canada** V100 GPUs across nine configurations; distillation lifted a weak student by 19.5 points on AgeDB (57.0% → 76.5%).

Denosing Diffusion Probabilistic Models — EECE 501

Dec 2023

- Initiated and led a three-person project; derived the **DDPM** training objective end to end and ran forward/reverse diffusion experiments with a noise-schedule comparison in **PyTorch**.

Ensemble stacking on CIFAR-10 — EECE 571T

May 2023

- Trained **LeNet**, **AlexNet**, **VGG** and **ResNet-18** as base learners plus a meta-network combining their predictions, evaluating every subset; the best three-model stack (88.7%) beat the best single model (85.6%) and included the weakest learner.

Feature extractors for few-shot learning — EECE 570

Apr 2023

- Trained **ResNet-18**, **DenseNet-121** and **ConvNeXt-T** from scratch as backbones for **Prototypical Networks** on CUB-200 under one episodic protocol (DenseNet 83.8% > ResNet 81.8% > ConvNeXt 78.6%); ablated two prototype-head modifications.

Denosing diffusion-weighted MRI — EECE 523

Dec 2022

- Built **block-DCT** thresholding and **PCA** denoisers from scratch in **NumPy** with an evaluation pipeline over three subjects and four b-values; **PCA** gave ~5 dB SNR gain vs ~1.7 dB for DCT.

Unsupervised brain-tumour segmentation — EECE 562

Dec 2022

- Implemented **Gaussian-mixture-model EM** from scratch in **PyTorch** over ~nine million voxels per study on BraTS 2018; reached Dice ≈ 0.9 on six of ten patients without any labels.

EDUCATION

M.Eng., Electrical & Computer Engineering — University of British Columbia, Vancouver

Sept 2022 – May 2024

GPA 87% · Focus: deep learning, computer vision, generative models, medical image analysis

B.Sc. (Honours), Computer Science & Mathematics — University of Victoria

Sept 2017 – May 2021

Graduated with distinction · GPA 7.53/9 · Honours thesis on off-policy reinforcement learning · Dean's List (2020)